17: Pipelining II (and Parallelism I)

ENGR 315: Hardware/Software CoDesign

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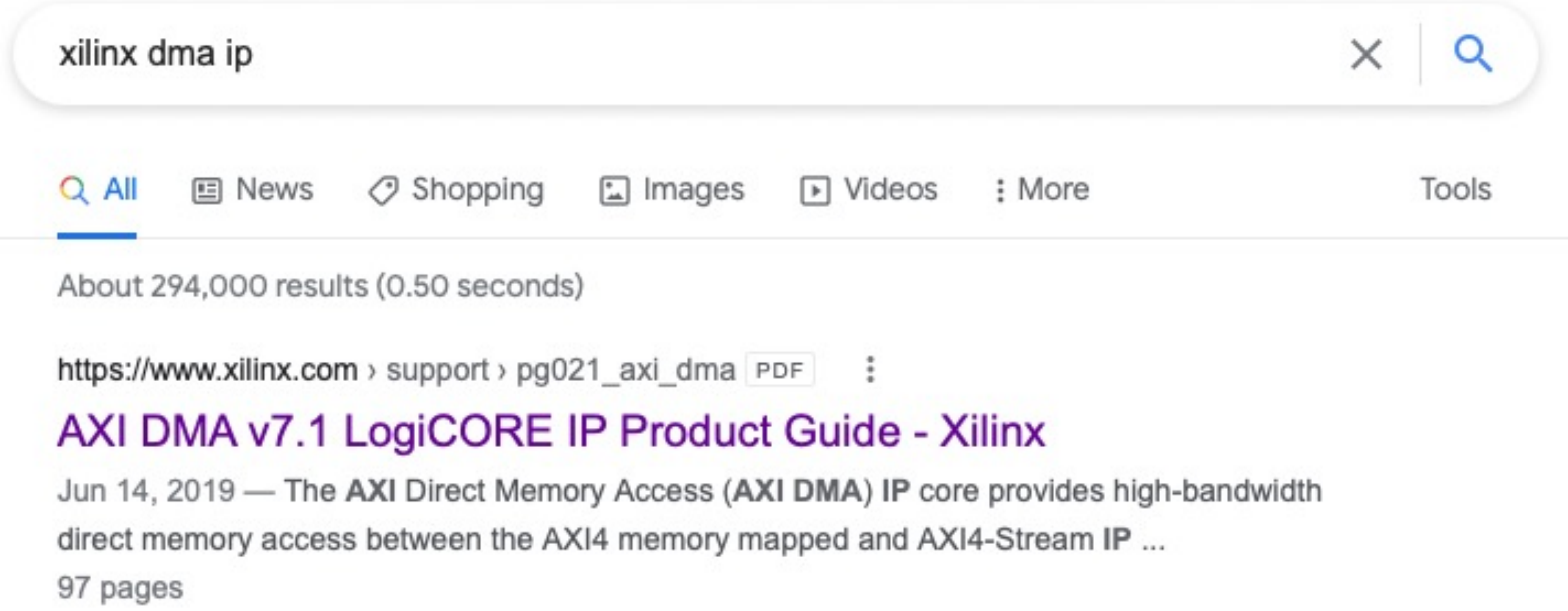
Some material taken from:
https://github.com/trekhleb/homemade-machine-learning/tree/master/homemade/neural_network
http://cs231n.github.io/neural-networks-1/

P7 – DMA from C



https://www.xilinx.com/support/documentation/ip_documentation/axi_dma/v7_1/pg021_axi_dma.pdf

P7 – DMA from C

Programming Sequence

Direct Register Mode (Simple DMA)

P7 – DMA from C

1. Start the MM2S channel running by setting the run/stop bit to 1 (MM2S_DMACR.RS = 1). The halted bit (DMASR.Halted) should deassert indicating the MM2S channel is running.

2. Skip

3. Write a valid source address to the MM2S_SA register.

4. Write the number of bytes to transfer in the MM2S_LENGTH register.

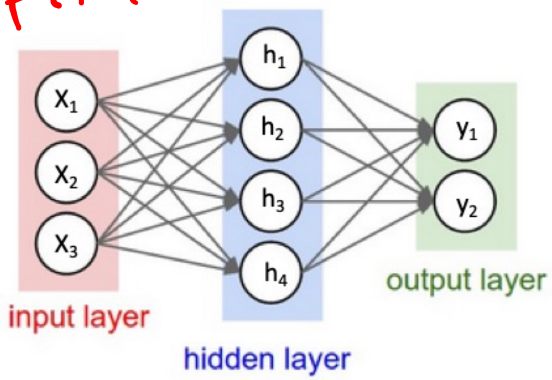
   The MM2S_LENGTH register must be written last.

5. Wait until MM2S_DMASR.Idle==1 for completion

P8+ are part of the "Design Challenge"

- Goal: Accelerate reference neural network
- Harder, more open-ended projects

Fix Pic



P8+ focus on Dot Product

```
# forward-pass of a 3-layer neural network:
f = lambda x: 1.0/(1.0 + np.exp(-x)) # activation function (use sigmoid)
x = np.random.randn(3, 1) # random input vector of three numbers (3x1)
h1 = f(np.dot(W1, x) + b1) # calculate first hidden layer activations (4x1)
h2 = f(np.dot(W2, h1) + b2) # calculate second hidden layer activations (4x1)
out = np.dot(W3, h2) + b3 # output neuron (1x1)
```

Matrix Multiplication (Dot Product)

$$\underbrace{\begin{bmatrix} 0.1 & 0.2 \end{bmatrix}}_{}^{\text{inputs}} \times \overbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}^{\text{weights}} =$$

$$= \begin{bmatrix} (0.1 \times 1 + 0.2 \times 4) & (0.1 \times 2 + 0.2 \times 5) & (0.1 \times 3 + 0.2 \times 6) \end{bmatrix}$$

(Answer)

$$= \begin{bmatrix} 0.9 & 1.2 & 1.5 \end{bmatrix}$$

Floating-Point math takes 8 cycles.

- Floating-Point is complicated.
- FMAC: a * b + c
- 8 cycles of complicated.
- How do we work around an 8 cycle latency?
- Pipelining!

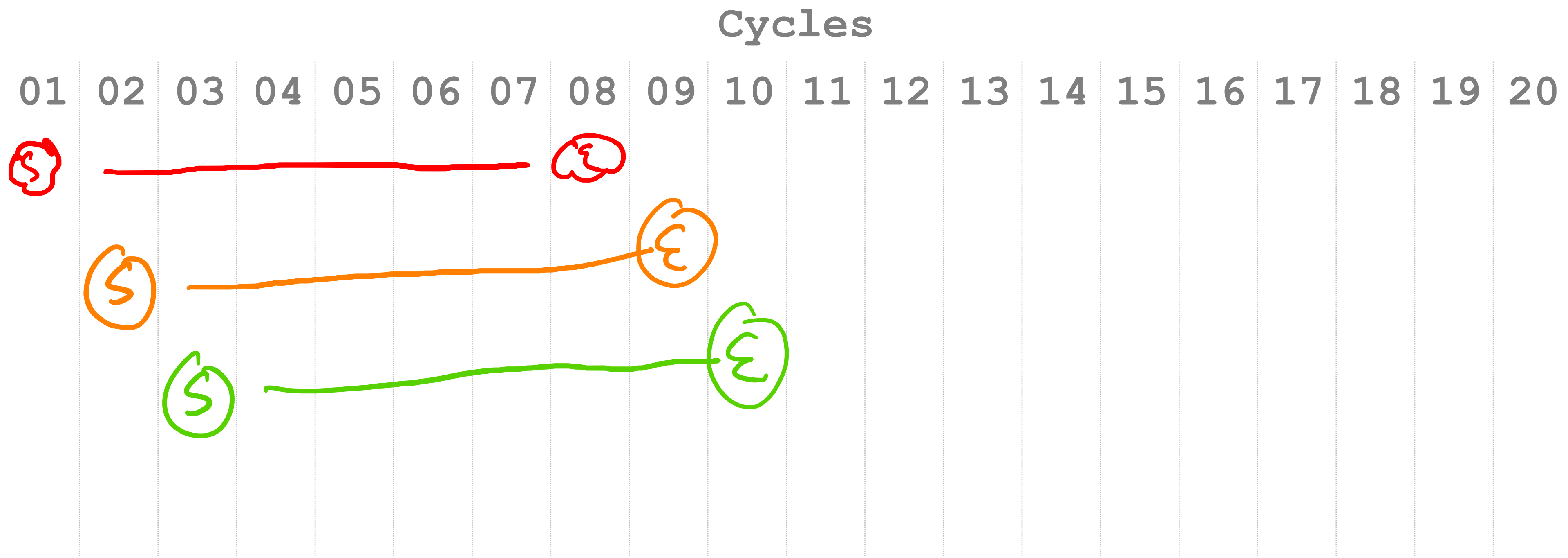
FMAC Pipelining

$$\begin{bmatrix} 0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$0.1 \cdot 1 + 0$

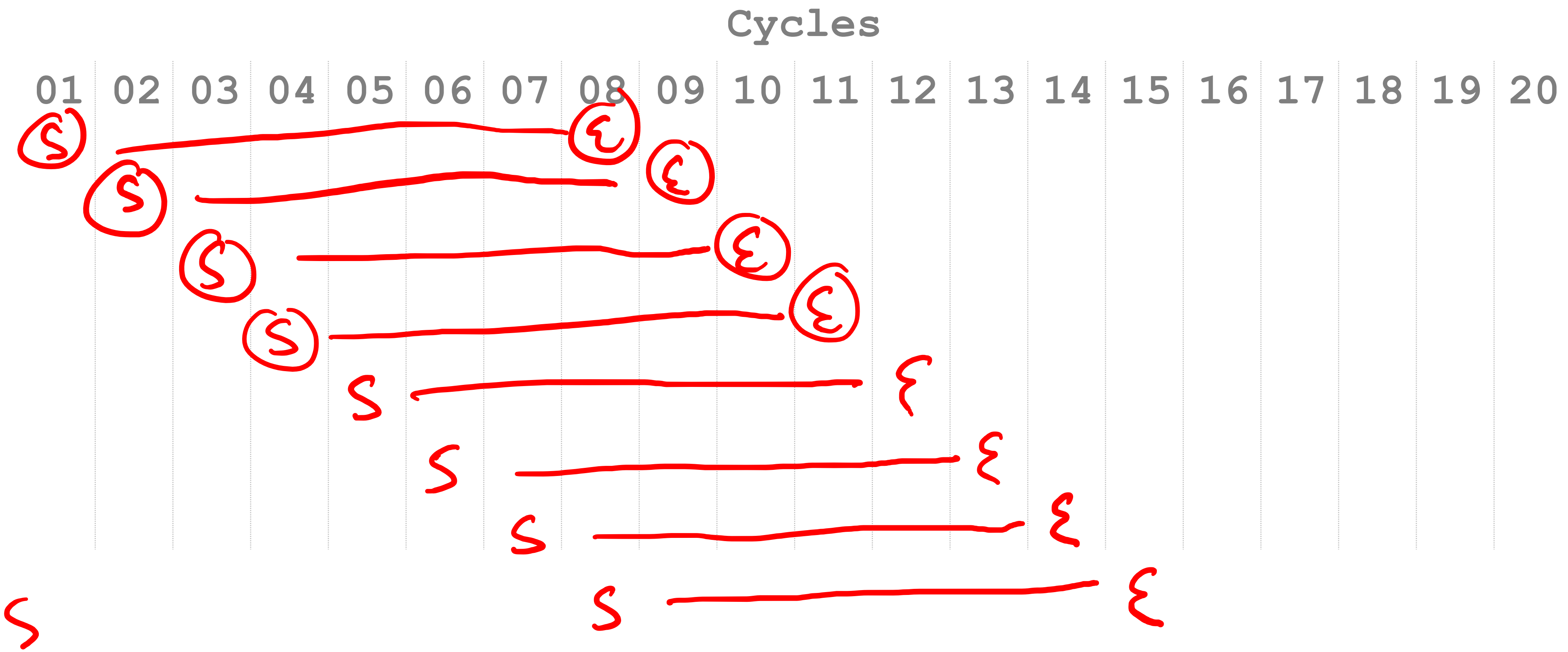
$0.1 \cdot 2 + 0$

$0.1 \cdot 3 + 0$



FMAC Pipelining

individual:
8 cycles



Average:
8 mults /
15 cycles

~2 cycles/mult.

~1/2 mult / cycle

Latency vs. Throughput

- **Latency**: How long does an individual operation take to complete?

individual mult: 8 cycles

- **Throughput**: How many operations can you complete per second (or per cycle)?

Average output per cycles:

~ 1 mult / cycle

Multiply-Accumulate Dot Computations

$$[0.1 \quad 0.2] \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = [0.9 \quad 1.2 \quad 1.5]$$

$$= [\underline{0} \quad \underline{0} \quad \underline{0}]$$

$$0.1 \cdot 1 + 0 \qquad 0.1 \cdot 2 + 0 \qquad 0.1 \cdot 3 + 0 = [\underline{0.1} \quad \underline{0.2} \quad \underline{0.3}]$$

$$0.2 \cdot 4 + 0.1 \qquad 0.2 \cdot 5 + 0.2 \qquad 0.2 \cdot 6 + 0.3 = [0.9 \quad 1.2 \quad 1.5]$$

Dependencies

$$\begin{bmatrix} 0.1 & 0.2 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

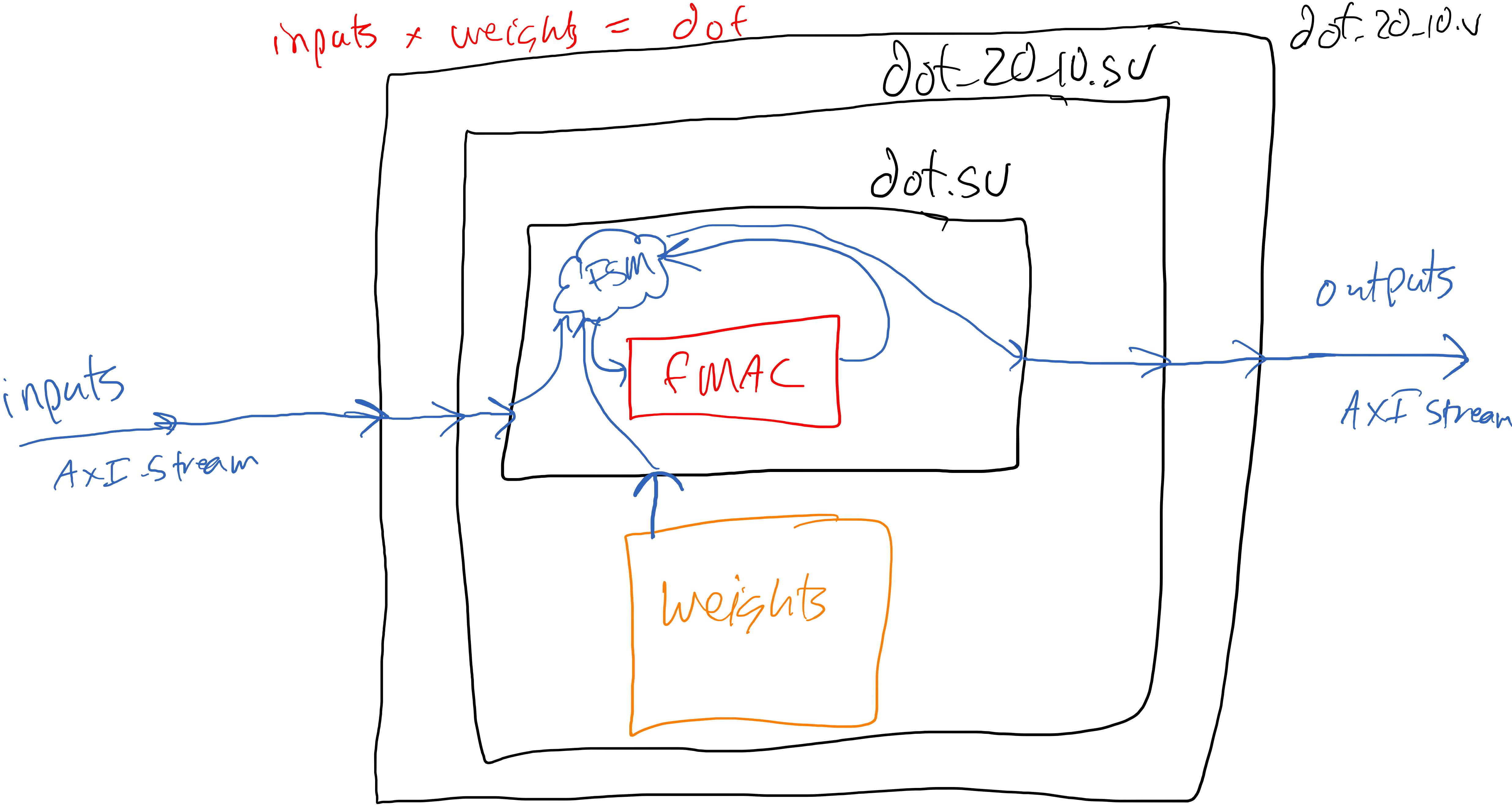
Cycles

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20

Dependencies on Pipelined FMAC

- Solution: Stall at the end of a row.
- “Drain” the pipeline.

- Restart new row
- “Refill” the pipeline



inputs × weights = dot
dot_20_10.v
dot_20_10.sv
dot.sv
FSM
fMAC
weights
inputs
AXI-Stream
outputs
AXI Stream

5.05

How many cycles *should* this take?

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$

8 cycles / multiply × 12 multiplies

= 96 cycles

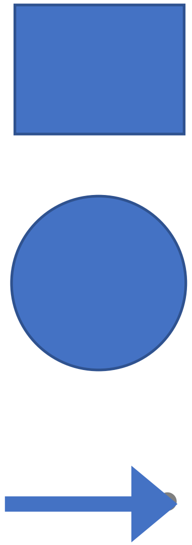
Data flow graph

- A data-flow graph is **a collection of arcs and nodes in which the nodes are either places where variables are assigned or used**, and the arcs show the relationship between the places where a variable is assigned and where the assigned value is subsequently used.

[ref]

Data Flow Graph

- Illustrates the dependencies between inputs and operations to produce an output



Inputs

Operations

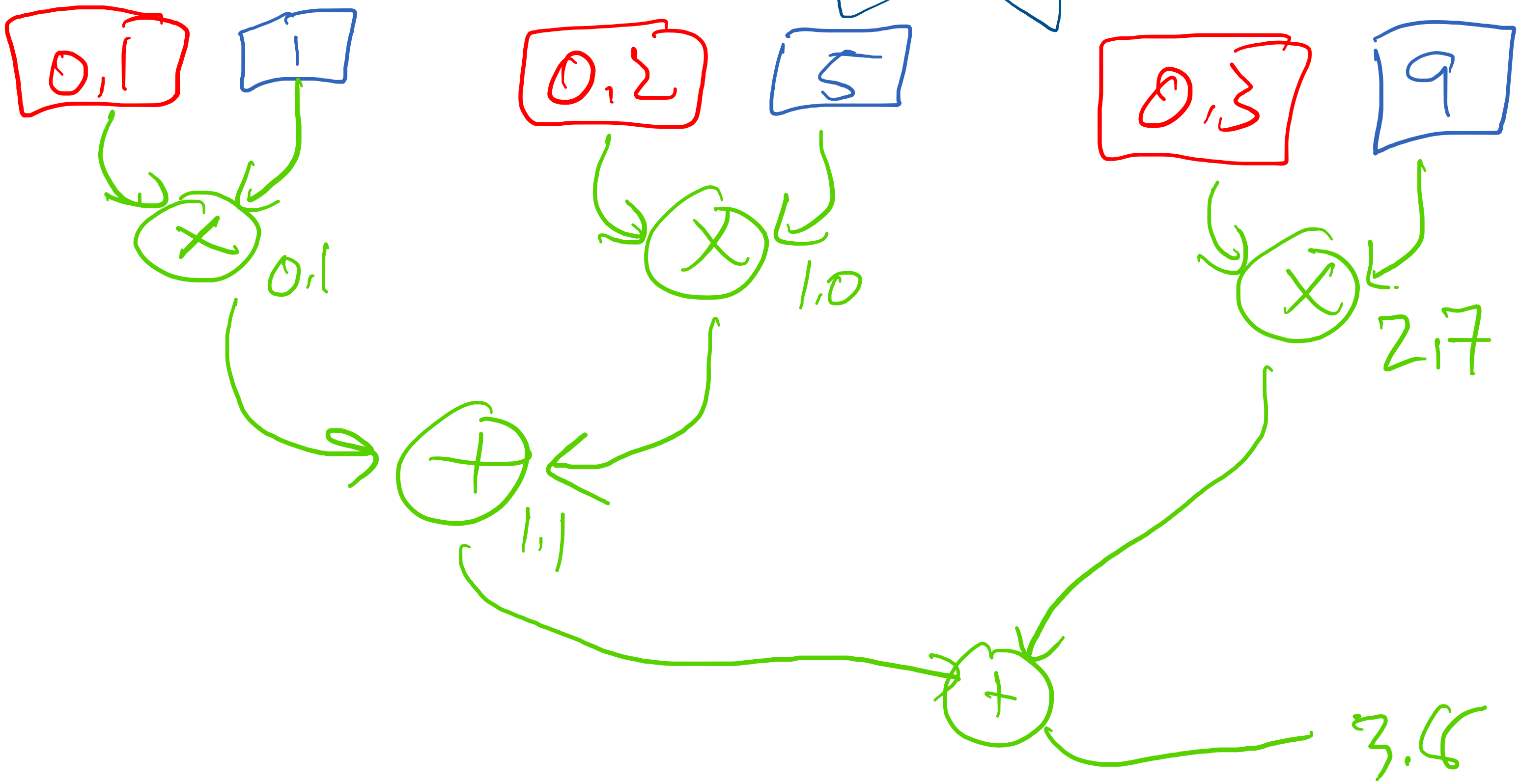
Dependencies

Data Flow Graph

* No MAC for Now

* No pipeline

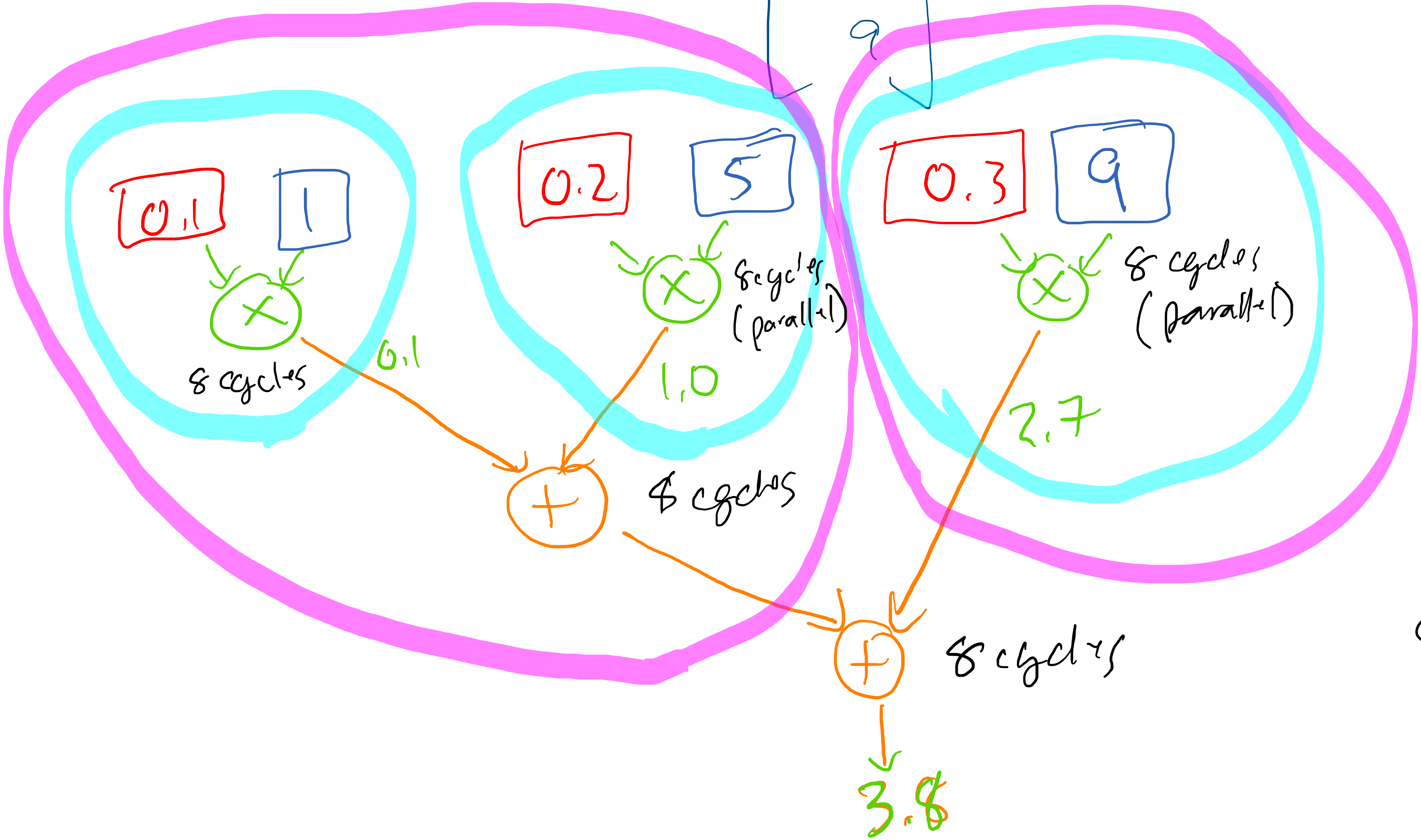
$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 3.8 \end{bmatrix}$$



Finding Parallelism

- Find some computation that doesn't depend on other computation's results.
- No arrows between blocks.
- Shared Inputs are OK.

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 3.8 \end{bmatrix}$$



Infinite Hardware

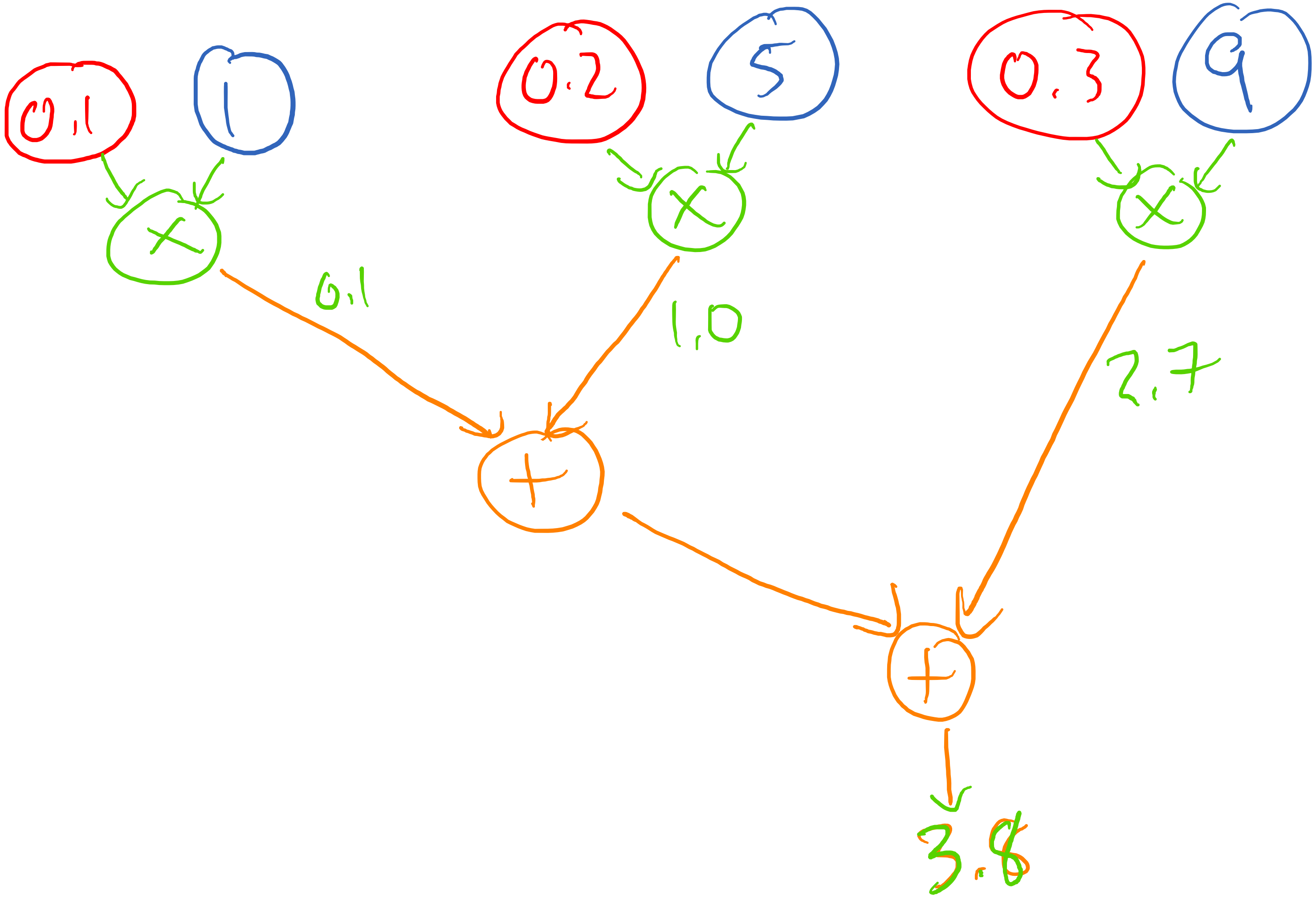
← 8 cycles

← 8 cycles

← 8 cycles

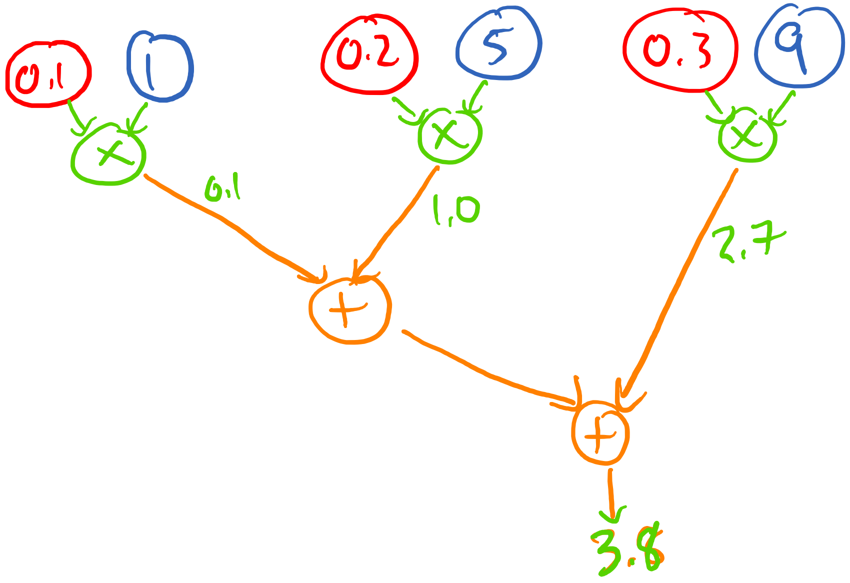
24 cycles

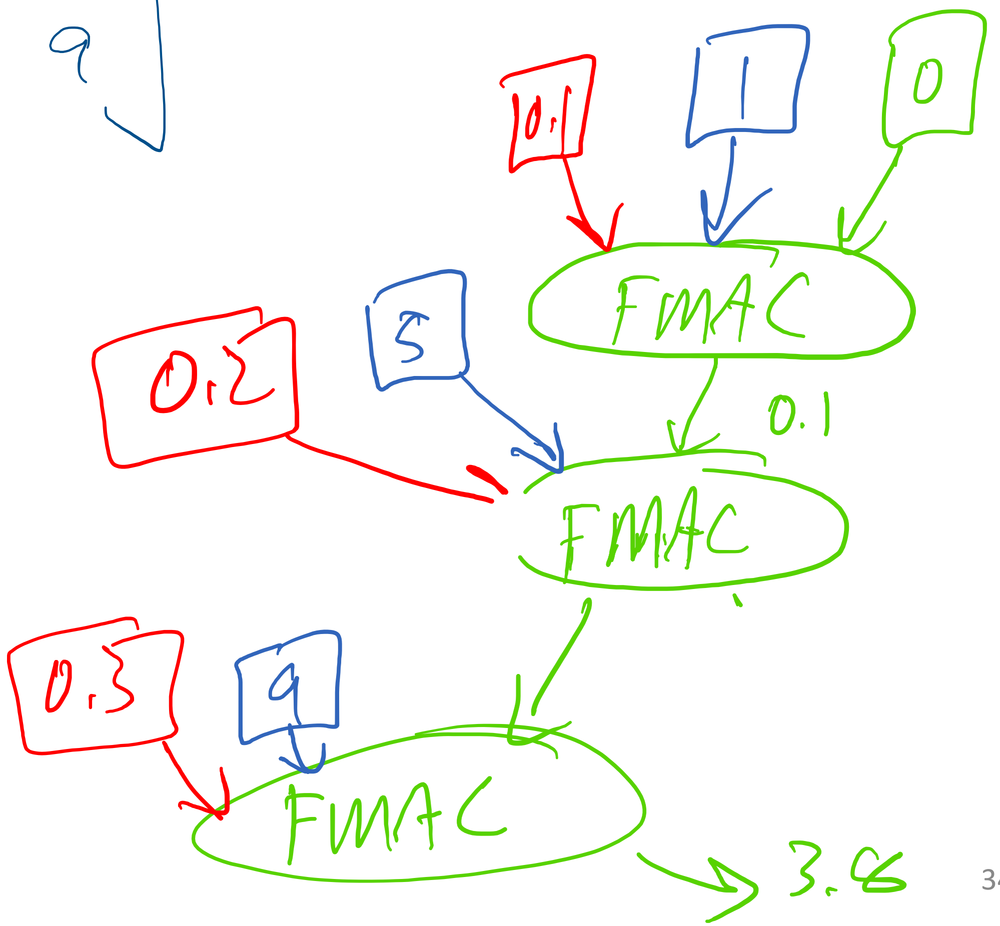
How many cycles *should* this take?



How many cycles *should* this take?

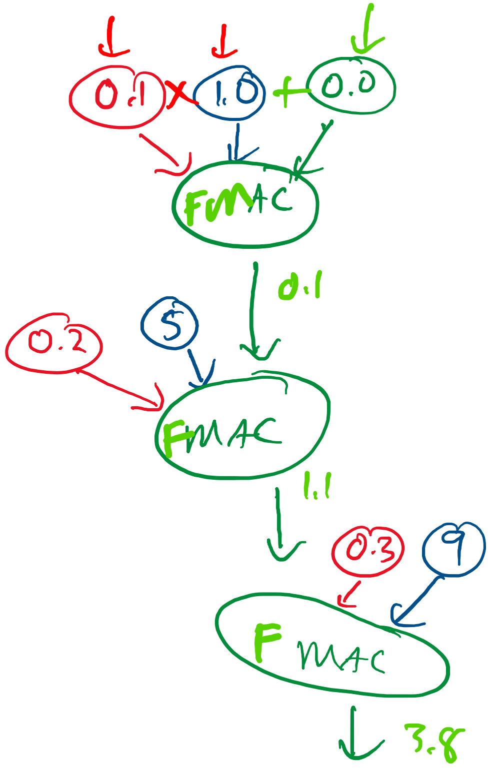
$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 3.8 \end{bmatrix}$$

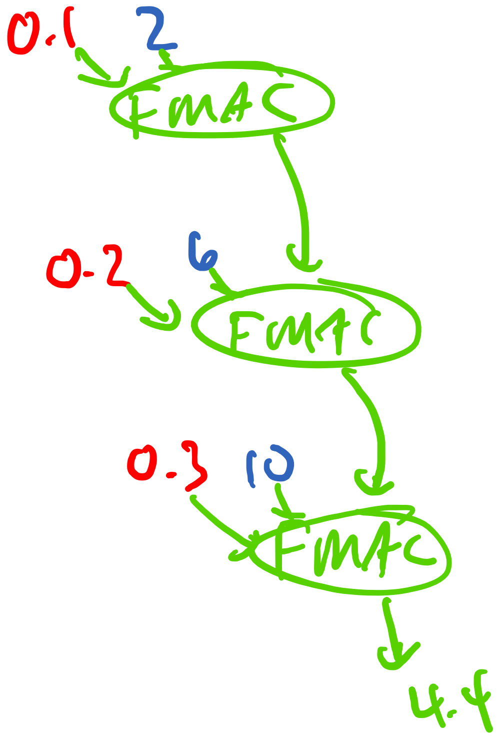




More Data Flow

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$



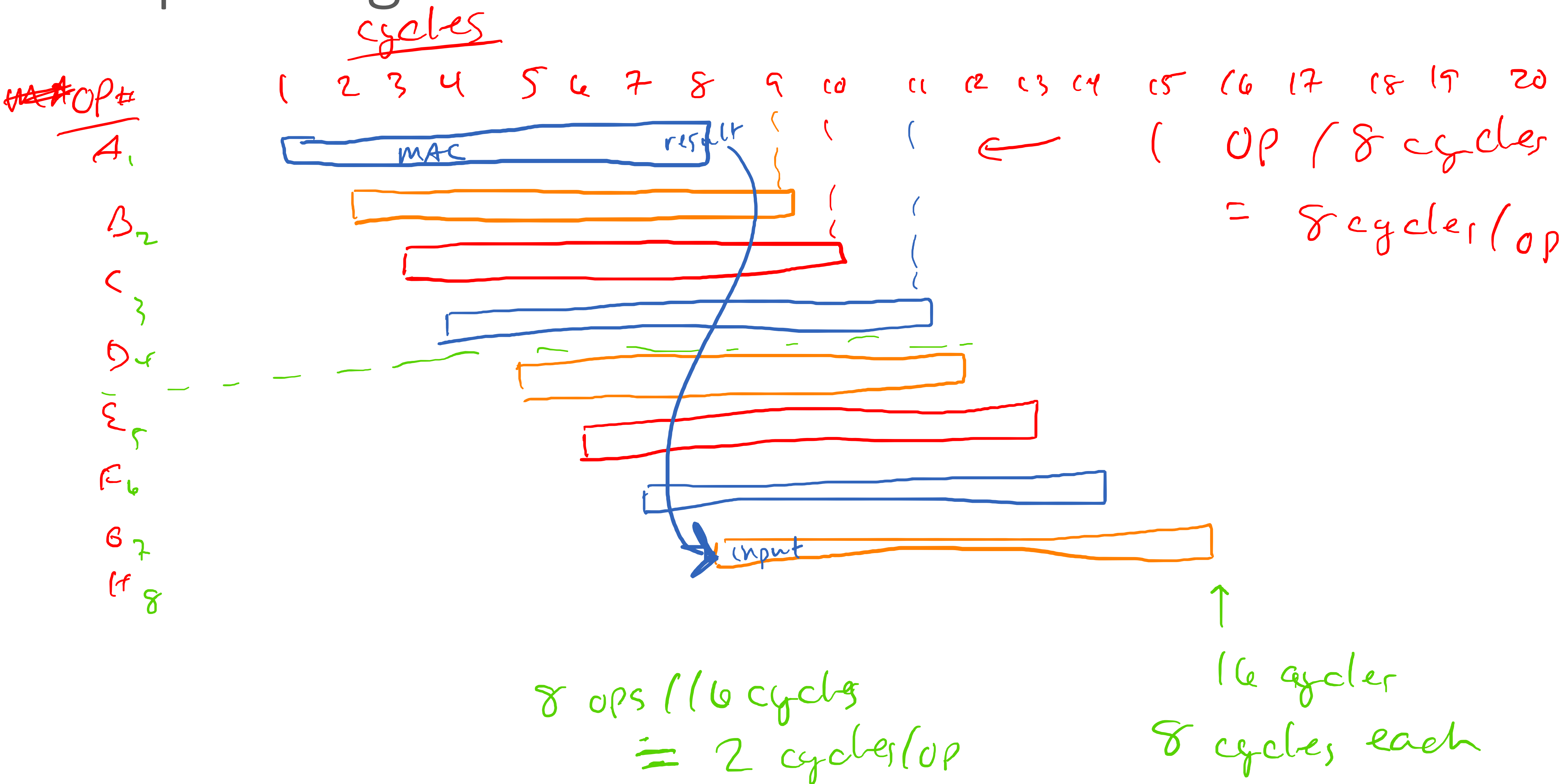


Pipelining

- FMAC takes 8 cycles for 1 value
- But can accept a new value every cycle.

- Latency: 8 cycles / value
- Throughput: 1 value / cycle

Pipelining


cycles
op#
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
A 1
B 2
C 3
D 4
E 5
F 6
G 7
H 8
MAC
result
input
1 op / 8 cycles
= 8 cycles/op
8 ops / 16 cycles
= 2 cycles/op
16 cycles
8 cycles each

$$\begin{bmatrix} \underline{0.1} & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ \underline{5} & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$

$$
\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix}
\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}
=
\begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}
$$

Only 1 or

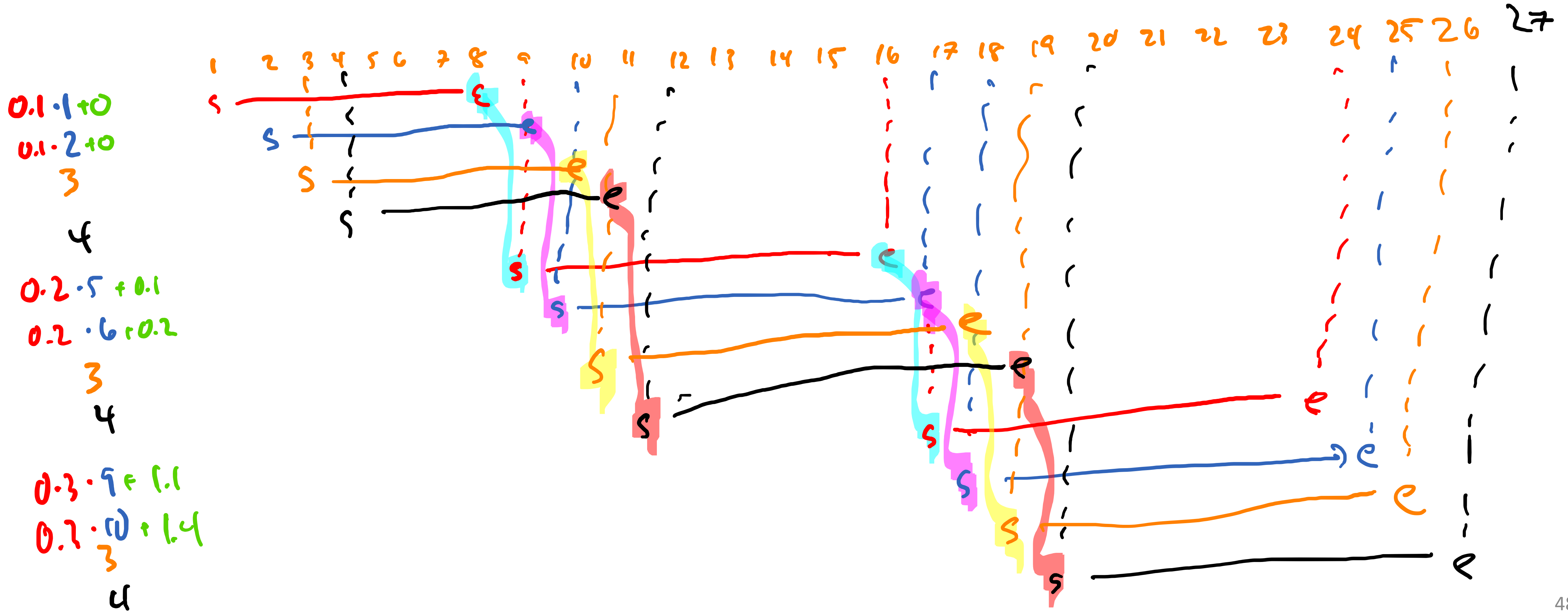
Infinite

	One Output	All 4 Outputs
Multiply & ~~Add~~ separate:	"higher"	
FMAC	24 cycles	24
Pipelined MAC	24 cycles	27 cycles
Maximum Parallel:	24	24

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$

	One Output	All 4 Outputs
Multiply & Add separate:	40 cycles	160 cycles
MAC	24 cycles	96 cycles
Pipelined MAC	24 cycles (+1 for extra rows)	33 cycles
Maximum Parallel:	24 cycles	24 cycles

Next Time: Divide and Conquer



$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ \hline 9 & 10 & 11 & 12 \end{array}\right] = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$0.1 \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix}$$

$$0.2 \cdot \begin{bmatrix} 5 & 6 & 7 & 8 \end{bmatrix} + \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} = \begin{bmatrix} 1.1 & 1.4 & 1.7 & 2.0 \end{bmatrix}$$

$$+ \begin{bmatrix} 2.7 & 3.0 & 3.3 & 3.6 \end{bmatrix}$$

$$0.3 \cdot \begin{bmatrix} 9 & 10 & 11 & 12 \end{bmatrix} + \cancel{\begin{bmatrix} 1.1 & 1.4 & 1.7 & 2.0 \end{bmatrix}} = \cancel{\begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}}$$

$$\begin{bmatrix} 3.8 & 4.4 & 5.0 & 5.6 \end{bmatrix}$$